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Detecting outliers in complex nonlinear systems controlled by predictive control strategy

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ABSTRACT

Detecting outliers in complicated nonlinear systems that are controlled by model predictive control is a significant work for engineering applications. Based on the features of data in practical systems, we propose a one-class classification ensemble method incorporating the notion of Feature Subspace with Bagging. Clustering and PCA (Principal Component Analysis) are integrated to obtain a more informative feature space, where Feature subspaces and bootstrap replications are implemented orderly to generate more accuracy and diverse base learners. A detector is constructed based on the above methodology, and a model updating strategy is also provided. By means of comparison with competitive methods, the effectiveness of the proposed detector has been verified.

1. Introduction

Nonlinearity is the most prominent feature of complex systems in multidisciplinary discipline, such as chemistry [1], metallurgy [2], aerospace [3], biology [4], physics [5] etc. Referring to the control problem of nonlinear systems, model predictive control (MPC) has been extensively researched. The basic control strategy in MPC is the selection of a set of future control moves (control horizon) and minimizing a cost function based on the desired output trajectory over a prediction horizon with chosen length [6]. This requires a reasonably accurate prediction model that captures the essential nonlinearities of the process under control, to predict multi-step ahead dynamic behavior. Typically, the prediction model is constructed once at the initial phase of a MPC scheme. Nevertheless, slow drifts in unmeasured disturbances and changes in process parameters will lead to significant mismatch in plant and model behavior as time progresses. Furthermore, a MPC scheme is usually designed under the assumption that sensors and actuators are free from faults, which is usually impossible for real-world systems, where soft faults, such as biases in sensors or actuators, are frequently encountered. Finally, sensors or actuators may also fail during operation, which results in loss of degrees of freedom for control. Such occurrences can lead to a significant degradation in the closed loop performance of the MPC scheme and may also lead to instability. We represent the structural schematic diagram of dynamic matrix control (DMC) [7] in Fig. 1, from which we could find that forecasting model plays a crucial role in DMC scheme. Then qualities of measurements of y_R, u, y_M, d, y will have great influence directly or indirectly on the final performance of DMC. As a result, we can conclude that the primary reason for these phenomena is that abnormal measurements (outliers) would participate directly or indirectly in the construction of prediction model, which leads to model-plant mismatch naturally. Therefore, timely detecting outliers and preventing them to construct the prediction model are extremely significant for MPC schemes of nonlinear systems. Following the definition of outlier by [8], we regard outliers as patterns in data that do not conform to a well-defined notion of normal behavior, which is usually represented by normal process data in the sense of practical nonlinear systems. For most the published work in the literatures, addressing outliers for nonlinear systems controlled by MPC scheme has rarely been considered, which motivates us for this study.

In this paper, we develop a novel direction that is referred to as outlier detection for nonlinear systems. According to the literature we have referred to, model predictive control was often used as an effective control strategy for those complex nonlinear systems. However, MPC has never been the only one, other methods like adaptive control has also been successfully applied in several nonlinear systems. In other words, our proposed outlier detection scheme may be appropriate for many nonlinear systems and more data-based control strategies including MPC. However, in this paper, our focus is *complex nonlinear systems controlled by MPC*. It is also worth stating that only the detection method

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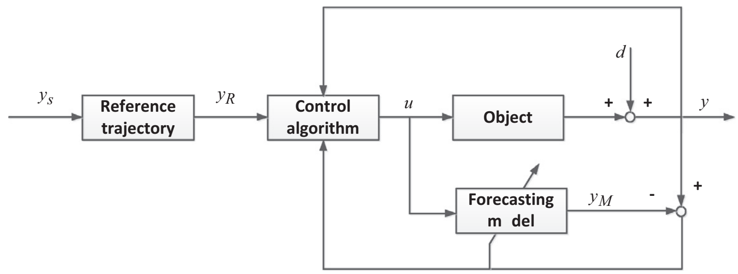

Fig. 1. Structural schematic diagram of dynamic matrix control (DMC).

is focused and the problem of control has not been researched here. Actually, the problem of outlier detection has drawn a significant amount of interests from artificial intelligence, data mining and machine learning in the past decade, reflecting in the publication of massive research papers, such as distance-based methods, density-based methods, classification-based methods, cluster-based methods [9] etc. In contrast to these traditional methods, in this paper, we investigate the characteristics of data and propose a novel and hybrid one-class classifier ensemble learning solution. Our method incorporates the notion of Bagging and Feature Subspace in order to deliver simultaneously accurate and diverse base learners. Our proposed feature selection method that integrates the thought of clustering with PCA can generate a new feature space, which is more concise and informative than the original feature space. Several feature subsets can be generated from this new feature space and more subsets can be further obtained via bootstrap sampling. With sub-models training on these subsets and an aggregation rule, unseen measurements can be grouped into two categories, namely normal samples and outliers. Finally, we propose a simple but appropriate strategy to update the detection method in order to capture the dynamics in process data.

The rest of this paper is structured as follows. The characteristic of data in practical chaotic systems is summarized in Section 2. Section 3 expands upon the proposed method. A series of experiments are carried out in Section 4. Finally, some conclusions are drawn in Section 5.

2. Features of data and method

In this section, we analyze the features of process data in practical nonlinear systems, based on which essential features of corresponding outlier detection method can be also be concluded.

(1) Data unlabeled. In most MPC schemes of nonlinear systems, data used for constructing the forecasting model are measurements of process data that are collected online. Labeling for these real-time measurements is a time-consuming task and even an impossible task.
(2) Noisy should be another prominent feature of data in practical systems. Due to the complicated operating conditions, noisy data is inevitable.
(3) High-dimensional data is increasingly pervasive in many areas with the development of advanced sensor technology. Due to the increasing scale of the nonlinear systems, requirements for control are becoming more rigorous. As a result, control performance can reach a high level only with more state variables participating into the control framework.
(4) In addition, inherent methodology of MPC also requests the detection to be implemented in real-time, which indicates that the detection phase should be fast enough.

As thus, we can conclude that a competent outlier detection method for practical nonlinear systems should: (1) be constructed with unlabeled data (unsupervised learning); (2) be robustness to noise in training set; (3) cope with high-dimensional data; (4) be implemented in real-time; (5) be fast enough. Although the algorithms for outlier detection can also be applied for fault diagnosis and detection (FDD), the ultimate objective is totally different from that of FDD. The aim of a FDD is similar with that of process monitoring, i.e. guaranteeing the product quality, while the aim of this paper is to provide convenience for process control. Both two processes are of great importance for industrial processes [10]. Furthermore, our proposed method can also facilitate the implementation of FDD like the scheme in [11].

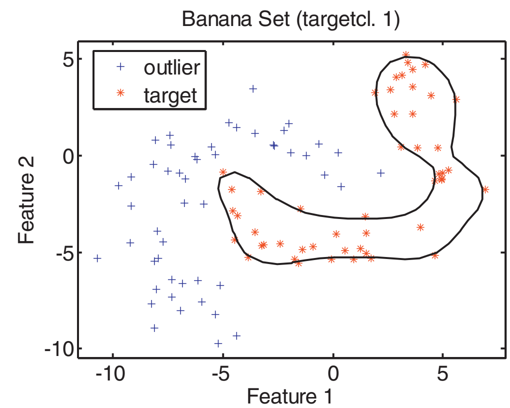

Fig. 2. An example of one-class classifier; "*" denotes target class, and "+" denotes outlier class; the solid line represents the trained boundary.

3. Methodology

According to the analysis in Section 2, we propose a hybrid and novel one-class classifier ensemble method, where the notion of Bagging [12] and Feature Subspace are combined to deal with the intricate data in practical nonlinear systems. The following subsections will expand upon methodology concerning our outlier detection method.

3.1. One-class classification

One-class classification (OCC) can be regarded as a special case of classification problem where the collection of counter-examples is hard even impossible, which is extremely appropriate for outlier detection under the condition of unlabeled data. In addition, the detection phase of classification-based method is simple and fast enough to be implemented in real-time. Generally, the task of OCC is to define a boundary around the target class (often normal examples), such that it accepts as much of the target objects as possible, while it minimizes the opportunity of accepting outlier objects. Fig. 2 illustrates a one-class classifier trained with a banana-shaped dataset. A number of OCC classifiers have been proposed in the literature and can be grouped into three categories.

3.1.1. Density-based methods

These methods are the most straightforward ones, which obtain a one-class classifier via estimating the density of the training data. For this type of methods, in general, a distribution should be assumed beforehand, such as Gaussian distribution, and a test which are referred to as discordancy test, is then available to test new objects. Among the most popular density-based methods for OCC issue, the mixture of Gaussian model and the Parzen density [13] are especially prominent.

3.1.2. Reconstruction-based methods

This type of methods was originally introduced as a tool for data modeling. Via utilizing prior knowledge about the data and making assumptions about the generating process, a model is chosen and fitted to the data. And the use of this type of methods for OCC issue is based on the idea that outliers deviate from this model. Of reconstruction-based methods, K-means, self-organizing map, auto-encoder and PCA [14] are the most popular ones.

3.1.3. Boundary-based methods

The main aim of this type of methods is to find the optimal size of volume enclosing given training points. Since these methods heavily rely on the distances between objects, they tend to be sensitive to scaling of the features. In addition, compared with the other two types of methods, since there is no need to estimate density or structure of the data, the required number of training examples is much smaller. One-class support vector machine (OCSVM) [15] and support vector data description (SVDD) [16] are the most popular ones among boundary-based methods.

3.2. Feature Subspace one-class classifier

Most methods' efficiency and effectiveness will be influenced somewhat when addressing high-dimensional data with noisy and redundant features. Inspired by the notion of Random Subspace (RS) [17], we propose a more efficient method to generate subspaces for constructing an ensemble one-class classifier. And this model will be mentioned henceforth as Feature Subspace_OCC (FS_OCC).

Specifically, we firstly employ a hierarchical clustering algorithm that uses Pearson correlation coefficient (PCC) as similarity measurement to partition the original features into M small groups.

$$P(i,j) = \frac{\sum_{k=1}^{N}(x_{ik}-\overline{x_i})\left(x_{jk}-\overline{x_j}\right)}{\sqrt{\sum_{k=1}^{N}\left((x_{ik}-\overline{x_i})\right)^2}\sqrt{\sum_{k=1}^{N}\left(\left(x_{jk}-\overline{x_j}\right)\right)^2}} \tag{1}$$

where x_{ik} is the value of feature i on the kth sample, $\overline{x_i}$ denotes the mean value of feature i on all training samples, and N is the number of training samples.

Then we employ PCA to determine the most primary feature in each small group so that we could obtain the new feature space (M-dimensional). Since PCA is an unsupervised dimension reduction method that determines the principal directions of the data distribution, it is appropriate to utilize PCA in OCC. Generally, one needs to construct the data covariance matrix and calculate its dominant eigenvectors in order to obtain these principal directions [18]. These eigenvectors will be the most informative among the vectors in the original data space, and are thus considered as the principal directions.

$$\min J(\mathrm{U}) = \sum_{i=1}^{n}\left\|(x_i-\mu)-\mathrm{U}\mathrm{U}^{\mathrm{T}}(x_i-\mu)\right\|^2 \tag{2}$$

Generally, the problem of (2) can be solved by deriving an eigenvalue decomposition problem of the covariance data matrix:

$$\sum_{A}\mathrm{U} = \mathrm{U}\Lambda \tag{3}$$

where

$$\sum_{A} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^{\mathrm{T}} \tag{4}$$

is the covariance matrix, μ is the global mean. Each column of U denotes an eigenvector of Σ_A, and the corresponding diagonal entry in Λ is the associated eigenvalues. Here we only need the largest eigenvalue and the corresponding feature. It is clear that the new feature space is constituted by features that are closely related with classification and approximatively non-redundant.

Finally, with the new feature space we employ one random project function P to generate several Feature Subspaces (d-dimensional).

$$P:\ R^M \to R^d \tag{5}$$

As thus, not only the diversity of base learners can be assured, but also the accuracy of each base learner can be improved compared with RS. As analyzed in [19], a necessary condition that an ensemble classifier outperforms its individual members is that base classifiers should be simultaneously accurate and diverse.

$$E = \overline{E} - \overline{A} \tag{6}$$

Predictably, the generated base learners by different Feature Subspaces can be more accurate than those in RS.

3.3. Feature_Subspace_Bagging one-class classifier

Khoshgoftaar has proved that the Bagging techniques generally outperform boosting in noisy data environments [20]. Hence, to cope with the feature of noisy, the notion of Bagging is introduced into the FS_OCC. And this model will be mentioned henceforth as FS_Bagging_OCC. Bagging, as an important and successful ensemble learning technique, incorporates bootstrap and aggregation, where bootstrap is a statistics approach using random sampling with replacement.

Actually, when introducing Bagging into FS_OCC, we have two ways depending on the orders of implementation of Bagging and FS. Specifically, one way is to bootstrap training samples firstly and generate Feature Subspaces on each subset secondly; the other way is to generate Feature Subspaces on all training samples firstly and generate subsets on each Feature Subspace secondly. As analyzed in [21], the main disadvantage of the first way is that more valuable information is lost since both dirty features and clear ones are deleted simultaneously. Therefore, the second way is preferred here, which can be demonstrated by Fig. 3. As can be seen from this schematic diagram, P Feature Subspaces are extracted from the training set S via the method proposed in Section 3.3. Then on any Feature Subspace, more than one bootstrap replication can be considered based on the problem at hand. T_{id_i} denotes the $d_{i\mathrm{th}}$ training subset for ith Feature Subspace. Finally, with regard to the problem of aggregation, $\sum_1^P d_i$ OCC model outputs need to be fused totally. Considering its simplicity and effectiveness for discrete output, majority voting (MV) is employed as the fusion method of FS_Bagging_OCC.

Predictably, the base learners of FS_Bagging_OCC can be more accurate than those in RS_OCC and more diverse than those in Bagging_OCC. In other words, it is expected that FS_Bagging_OCC can outperform RS_OCC and Bagging_OCC in OCC problem according to formula (6).

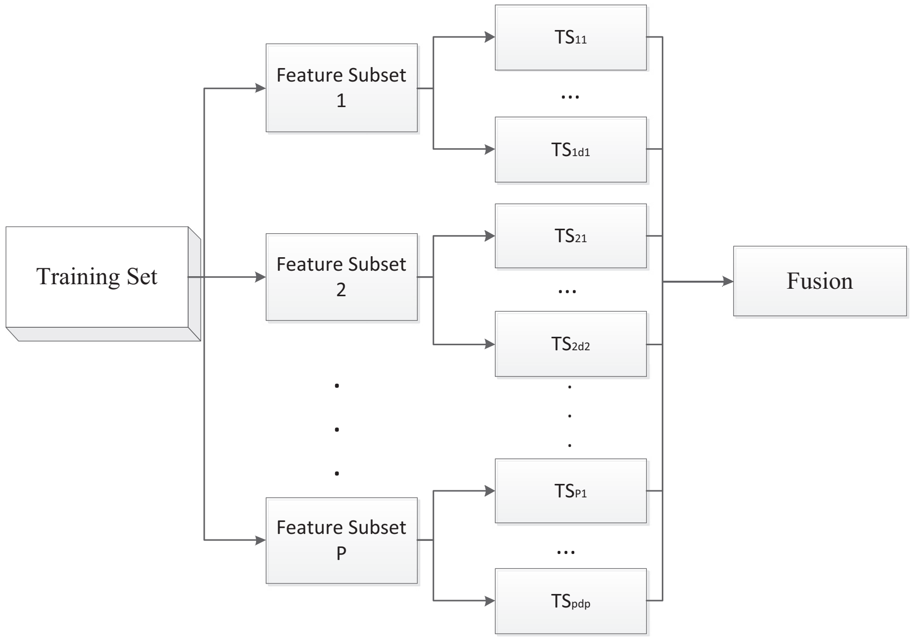

Fig. 3. The schematic diagram of FS_Bagging_OCC.

Input: Training set $\mathbf{S} \in \boldsymbol{R}^{\boldsymbol{K}}$, the number of clusters M, the dimension of feature subspace $\boldsymbol{d}$, the number of feature subspaces P, resampling rate $\boldsymbol{\alpha}$, the number of bootstrap replications $\boldsymbol{d_i}$; Testing sample x.

Process:

1. Calculate PCC matrix PC; /* PC_{ij} denotes the PCC between feature i and j.
2. Clustering (PC)→$\boldsymbol{cluster_i}$ $\mathbf{1} \leq \boldsymbol{i} \leq \boldsymbol{M}$;
3. **For** i=1:M
4. {
5. PCA($\boldsymbol{cluster_i}$)→$\boldsymbol{v_i}$
6. }
7. $[\boldsymbol{v_1}, \boldsymbol{v_2}, \dots, \boldsymbol{v_M}]$→$\boldsymbol{R^M}$; /* New feature space.
8. $\boldsymbol{f}(\boldsymbol{R^M} \rightarrow \boldsymbol{R^d}) \rightarrow \boldsymbol{FS_j}$, $\mathbf{1} \leq \boldsymbol{j} \leq \boldsymbol{T}$; /* Random projection.
9. **For** j=1:T
10. {
11. Bootstrap sampling ($\boldsymbol{FS_j}$)→ $\boldsymbol{TS_{j1}}, \dots, \boldsymbol{TS_{jd_j}}$;
12. }
13. OCC ($\boldsymbol{TS_m}$)→ $\boldsymbol{SM_m}$, $\mathbf{1} \leq \boldsymbol{m} \leq \sum_{\boldsymbol{j}} \boldsymbol{d_j}$; /* Construct sub-models.
14. MV ($\boldsymbol{SM_m}$)→ $\boldsymbol{EM}$; /* Construct ensemble model by majority voting.
15. $\boldsymbol{EM} \rightarrow \boldsymbol{c}(\boldsymbol{x})$.

Output: $\boldsymbol{c}(\boldsymbol{x})$; /* Label of testing sample.

Fig. 4. The pseudo-code description of FBO detector.

3.4. FS_Bagging_OCC detector

Based on the methodology proposed in Section 3.3, we develop an outlier detector that will be mentioned henceforth as "FBO detector", which stands for **FS_Bagging_OCC detector**. Figs. 4 and 5 provide the detailed pseudo-code description and schematic diagram of FBO detector, respectively. It can be found that FBO detector comprises three components:

(1) **Online training**. This is the most important phase of the detector since its performance determines the final result of detection directly. With the measurements collected in real-time, we use FS_Bagging_OCC to train a reference model, which will be saved and used for detecting the subsequent measurements.

(2) **Online detection**. With the reference model constructed in the training phase, new arriving measurement will be detected. Furthermore, normal samples and outliers are stored in two independent buffers, which will be used in the next phase.

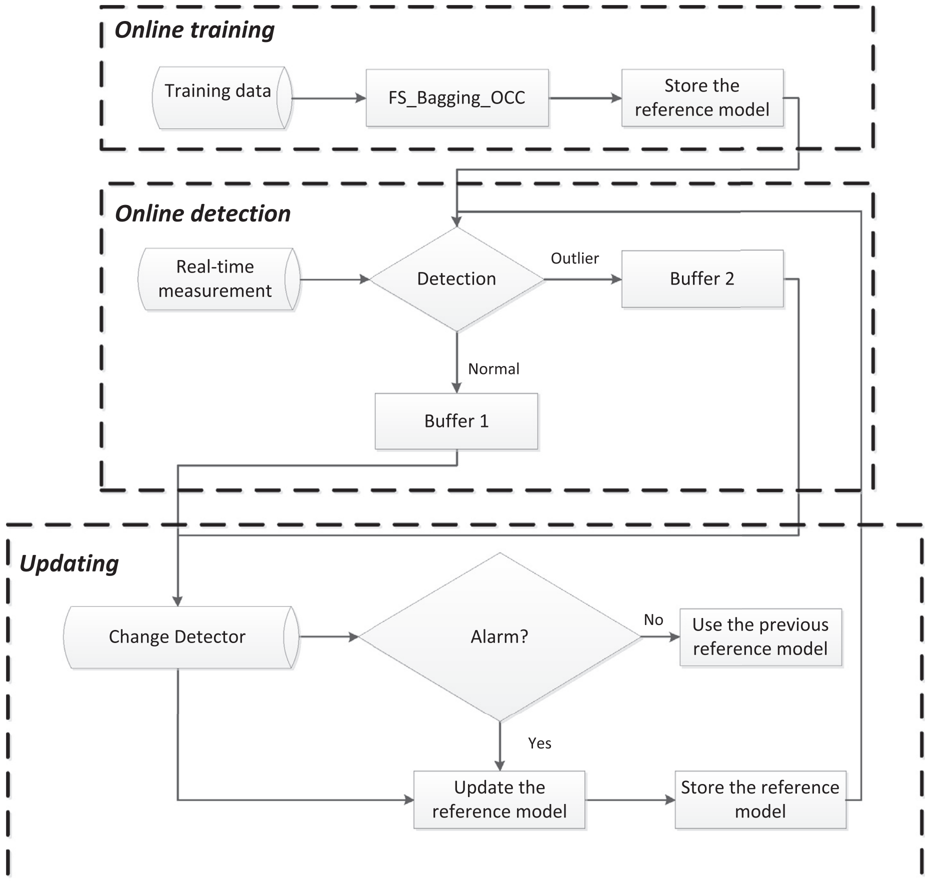

Fig. 5. The schematic diagram of FBO detector.

(3) **Model updating**. In order to make FBO detector adaptive to the dynamics of data distribution, we propose a simple adaptation strategy to update the detector automatically. Firstly, a change detector is designed to determine when to update FBO detector. The change detector contains two indicators: (a) does the number of measurements in any buffer reach the upper limit; (b) do outliers occur in group. If any indicator raises an alarm, then we will use some newer measurement to retrain the reference model.

These three phases work systematically and constitute a closed loop so that FBO detector is able to identify outliers as many as possible while rejecting normal samples as few as possible.

4. Experiments and analysis

In this section, we carry out a series of experiments on several benchmark datasets from UCI database and two real complex nonlinear systems to investigate the performance of FBO detector.

4.1. Datasets

A brief description of datasets used in our experiments is provided in Table 1. These two datasets are taken from two real systems, namely electric arc furnace (EAF) system and transonic wind tunnel (TWT) system, respectively. EAF systems are suitable devices to melt scrap and direct reduced iron for steel production and have been extensively used in many countries for refining quality steel for industry [22]. Strong nonlinearity is the most prominent feature in EAF systems, and MPC schemes have also been applied into the control framework. TWT systems are used for testing scale models, mostly of airplanes, in the speed region of 0 to Mach 1.3. Also strong nonlinearity is highlighted in TWT systems and it has been verified that predictive controllers for the Mach number have significantly better performance than PID controllers and the key issue in predictive control is the prediction model for Mach number [3].

Table 1
Descriptions of dataset extracted from two real-world nonlinear systems.

Dataset	Number of samples	Normal:outliers	Number of features
EAF	2000	10:1	17
TWT	2000	10:1	9

Actually, synthetic outliers and noise are randomly injected into both the training set and testing set. As is shown in Table 1, the ratio of normal samples to outliers is 10:1 for both datasets.

4.2. Baselines

To demonstrate the potential and necessity of FBO detector for outlier detection in nonlinear systems on high-dimensional and noisy data, we compare our method with the following baselines:

(1) Single OCC (SO) model.
OCC models have been widely used for outlier detection in many areas. Thus, we compare our method with single best OCC model (including PCA, SVDD, KNN, and MST) in high-dimensional and noisy environment.
(2) RS_OCC (RO) model.
Random subspace has been extensively used to construct an ensemble model and performs well for high-dimensional data.

Table 2
Confusion matrix of two-class classification problem.

		Actual label	
		Target class	Negative class
Predicted label	Target class	True positive (TP)	False positive (FP)
	Negative class	False negative (FN)	True negative (TN)

Thus, we compare our method with RS_OCC high-dimensional and noisy environment.

(3) Bagging_OCC (BO) model
Ensemble model constructed with Bagging technique usually perform well for noisy data. Thus, we compare our method with Bagging_OCC high-dimensional and noisy environment.

(4) RS_Bagging_OCC (RBO)
To emphasize the effectiveness of our proposed feature selection method based on clustering and PCA for the ensemble learning. We compare our detector with RS_Bagging_OCC.

4.3. Metrics

Referring to the evaluation criteria for methods of classification, many metrics have been developed, such as overall accuracy, precision, recall, geometric mean of accuracies (*G*-mean), *F*-measure, etc. In addition to these metrics for only one specific threshold of one-class classifier, there are also some metrics that can evaluate the average performance of a classifier for various thresholds, such as ROC-curve and PR-curve. As the research community continues to develop a greater number of intricate and promising learning algorithms, it becomes paramount to have standardized evaluation metrics to properly assess the effectiveness of such algorithms. In this paper, we choose four metrics that are more suitable for data in practical systems (data imbalance), namely *G*-mean, *F*-measure, ROC-curve.

Prior to the introductions of these metrics, a representation of classification performance is formulated by a confusion matrix as illustrated in Table 2. It is worthy to state that outlier samples belong to positive class and normal samples belong to negative class. Then we can formulate G-mean as:

$$G\text{−mean} = \sqrt{\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \times \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}} \tag{7}$$

This metric evaluates the degree of inductive bias in terms of a ratio of positive accuracy and negative accuracy.

F-measure can be formulated as:

$$F\text{−measure} = \frac{\left(1+\beta^2\right) \times \mathrm{Recall} \times \mathrm{Precision}}{\beta^2 \times \mathrm{Recall} + \mathrm{Precision}} \tag{8}$$

where β is a coefficient to adjust the relative importance of precision versus recall (usually, $\beta = 1$).

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{9}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{10}$$

F-measure combining recall and precision as a measure could provide more insight into the functionality of a classifier than the accuracy metric.

Referring to the ROC-curve, both true positives rate and false positive rate are used to evaluate the average performance of a classifier by providing a visual representation of the relative trade-off between the two metrics. Then, AUC is an evaluation criterion that uses the area under the ROC-curve [23].

Table 3
Performance evaluation of different methods for EAF dataset.

Method	Metrics		
	G-mean	*F*-measure	AUC
Single OCC	0.608	0.498	0.842
RS_OCC	0.647	0.602	0.889
Bagging_OCC	0.639	0.593	0.903
RS_Bagging_OCC	0.701	0.693	0.945
FS_Bagging_OCC	**0.762**	**0.768**	**0.953**

Table 4
Performance evaluation of different methods for TWT dataset.

Method	Metrics		
	G-mean	*F*-measure	AUC
Single OCC	0.761	0.690	0.931
RS_OCC	0.771	0.726	0.943
Bagging_OCC	0.793	0.747	0.950
RS_Bagging_OCC	0.833	0.781	0.969
FS_Bagging_OCC	**0.839**	**0.792**	**0.981**

4.4. Setup of FBO detector

(1) FBO detector is designed to work with one-class classifiers, and the choice of the type of classifier is left to the end-user of the proposed method. Comparing results by different OCC models is not the objective of this paper. Thus, considering its simplicity and effectiveness, SVDD is employed as the base learner of FBO detector.
(2) Then with regard to the selection of hierarchical clustering method, agglomerative nesting (AGNES) has been chosen eventually due to its simplicity. For determination of the number of clusters, an entropy-based method is employed to accomplish this work. It is necessary to claim that this criterion may be not the best solution for estimating the number of clusters. However, the influence is not significant from the perspective of ensemble learning.
(3) For all feature subsets generated from the new feature space, the re-sampling rate is set as the same value. And the number of bootstrap replications generated from each feature subset is also set as the same value.

Considering the limited space of paper, the calculation processes of these factors are omitted.

4.5. Results and analysis

Tables 3, 4 and Figs. 6, 7 report the results of different methods on two real-world datasets. From results in these two tables, we observe as follows:

(1) Ensemble methods usually outperform the corresponding single model. This phenomenon is more obvious for EAF dataset compared with TWT dataset. The reason may be that EAF dataset is more intricate with respect to both dimension and noisy level. This also indicates that ensemble models are more suitable for complicated environment.
(2) Compared with RS_OCC and Bagging_OCC, RS_Bagging_OCC obtains better performance for both datasets. This indicates that models combining RS with Bagging are more competitive than both RS and Bagging when dealing with complicated datasets.
(3) Compared with RS_Bagging_OCC, FS_Bagging_OCC obtains better performance for both datasets, which indicates that our proposed feature selection method based on clustering and PCA is more effective than RS. However, the improvement for TWT

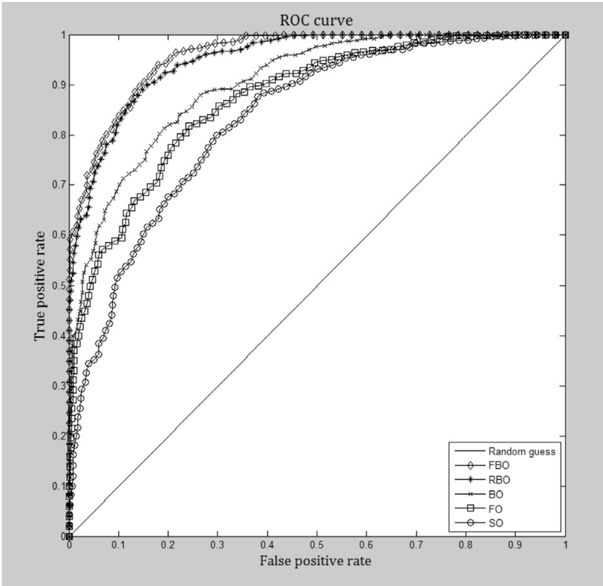

Fig. 6. Comparison of ROC-curve for different methods in terms of EAF dataset.

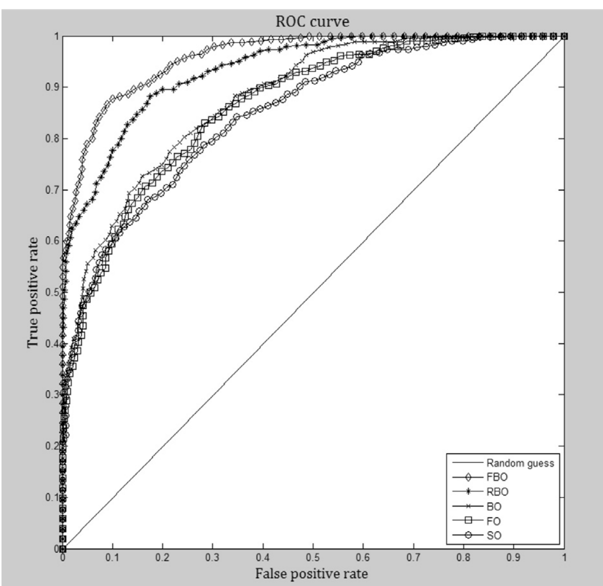

Fig. 7. Comparison of ROC-curve for different methods in terms of TWT dataset.

dataset is more limited than that for EAF dataset. The reason is that the dimension of TWT dataset is so low that the ability of our proposed Feature Subspace method cannot be developed sufficiently. This indicates that our proposed method is not suitable for processing low-dimensional data since the proposed FS method can help yield enough diverse training subsets and promote the efficiency of ensemble learning only for high-dimensional data.

(4) Note that for the actual application of a one-class classifier a specific threshold has to be chosen, which indicates that only a single point of the ROC-curve is used. Therefore, it can happen that for a specific threshold a one-class classifier with a lower AUC might be preferred over another classifier with a higher AUC. It just means that for that specific threshold, the fraction of false positive is smaller for the first classifier than the second classifier. This is the reason why the differences among different methods are small in terms of G-mean and F-measure metrics.

It is natural that the time complexity of FBO detector is relatively higher than that of other four methods. Nevertheless, this increment is so slight that can be ignored in practical engineering.

5. Conclusions

This paper proposes a novel and practical direction, namely outlier detection dedicates to complicated nonlinear systems controlled by MPC schemes. Our aim is to provide high-quality measurements for the construction of prediction model in MPC schemes. After sufficient analysis of the characteristics of measurements in general complicated nonlinear systems, we conclude that the outlier detector should: (1) be constructed with unlabeled data (unsupervised learning); (2) be robustness to noise in training set; (3) cope with high-dimensional data;(4) be implemented in real-time; (5) be fast enough. Therefore, we propose an ensemble OCC model that incorporates the notion of Feature Subspace and Bagging. A novel feature selection method that combines clustering with PCA is proposed to generate more informative feature space for high-dimensional data. The introduction of Bagging can help enhancing the robustness to noise in the training set. Also, a simple model updating strategy is proposed to improve the adaptivity of the detector. Via a series of experiments, the effectiveness of FBO detector has been verified. The results indicate that our proposed method is more suitable for high-dimensional and noisy data, which is just the characteristic of data in real-world nonlinear systems. Actually, the performance of detection method can be further improved. By employing both positive and negative samples collected online, a binary classifier could be constructed. Since more data information is utilized, a binary classifier could achieve better results. This is just our further research direction.

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